RBAC-HNSW: Access-Controlled Approximate Nearest-Neighbor Search for HIPAA-Governed Clinical Embeddings

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ABSTRACT

Clinical AI systems store millions of high-dimensional vector embeddings of patient records and must answer similarity search queries in real time while enforcing HIPAA access control. Post-filtering (retrieve candidates, then discard unauthorized ones) collapses to near-zero recall when access is restricted to $\leq$ 1% of records. Brute-force pre-filtering is exact but scales poorly. Multi-index architectures (e.g., SIEVE) consume 8–64× more memory for realistic role hierarchies.

We present **RBAC-HNSW**: a single modification to the HNSW graph-search algorithm that evaluates a 64-bit RBAC bitmask gate *before* the distance computation (≈1 ns vs. ≈30 ns) and routes through denied nodes to preserve graph connectivity. Memory overhead is 0.24%. At $N = 200{,}000$ vectors, QPS vs. Recall@10 Pareto curves show that post-filter recall collapses to zero at strict selectivity (0.02%) while RBAC-HNSW maintains Recall@10 ≥ 0.99 across all ef values, and dominates the Pareto frontier at every selectivity level tested.

CCS CONCEPTS

• **Information systems → Database query processing**; *Data privacy*.

KEYWORDS

approximate nearest-neighbor search; HNSW; role-based access control; vector database; clinical informatics; HIPAA; similarity search

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The source code, data, and/or other artifacts have been made available at https://github.com/patriotsbreeze/Similarity-RBAC.

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1 INTRODUCTION

The deployment of transformer-based language models in clinical medicine has produced a new infrastructure challenge: *access-controlled approximate nearest-neighbor (ANN) search* over protected health information (PHI). Consider a hospital AI system storing BioBERT [5] embeddings of one million clinical notes (768-dimensional float vectors) that must answer each query in under 10 ms while enforcing HIPAA: an Oncology physician must not see Psychiatry notes; a researcher may access only records with explicit consent.

The selectivity problem. In Role-Based Access Control [3] (RBAC), each user holds a *permission mask* determining which records they may read. *Selectivity* is the fraction of the database accessible to a query. In realistic hospitals, selectivity ranges from ≈80% (broad-access administrator) down to ≈0.1% (researcher querying HIV-positive trial data). No existing ANN index handles this full range efficiently.

- **Post-filtering.** Standard HNSW [6] retrieves candidates, then discards unauthorized ones. At 0.02% selectivity a beam of $ef =$ 800 yields $800 \times 0.0002 = 0.16$ accessible results on average—recall collapses toward zero.
- **Pre-filtering (brute-force).** Exact scan of the accessible subset: $O(|\text{accessible}| \times d)$ per query; at 40% selectivity over 1M vectors this gives single-digit QPS.
- **Multi-index (SIEVE [10]).** One HNSW index per role. Good recall and QPS, but memory scales with |roles|: 64 roles ⇒ 64× baseline (≈213 GB for 1M, 768-dim vectors; Section 4).

Related work. FilteredDiskANN [4] builds per-label entry-point structures for disk-based filtered search. ACORN [8] augments neighbor lists with predicate-satisfying nodes; our approach requires no graph augmentation. VBASE [9] observes that beam traversal can progress through filter-failing nodes—an insight we formalize for RBAC with bitmask semantics and zero distance overhead for denied nodes.

Contribution: RBAC-HNSW. We present a *single modification* to the HNSW greedy search: a 64-bit bitmask gate evaluated *before* the vector distance computation (≈1 ns vs. ≈30 ns). Denied nodes are not pruned from graph traversal—their neighbor lists are still used to *route* the beam toward accessible regions, preventing "connectivity deserts." Memory overhead: 8 bytes per vector (= **0.24**%). Open-source: https://github.com/patriotsbreeze/Similarity-RBAC.

Algorithm 1 RBAC-HNSW Greedy Search (Layer 0)

Require: Graph G, query $\mathbf{q}$, query mask q_m, result count k, beam *ef*
Ensure: Top-k accessible nearest neighbors
1: $results \leftarrow \emptyset$; $routing \leftarrow \{e\}$; $visited \leftarrow \{e\}$
2: **while** $routing \neq \emptyset$ **and** $|results| < ef$ **do**
3: $\quad (d_c, c) \leftarrow \textsc{Pop}(routing)$
4: $\quad$ **for all** $n \in \textsc{Neighbors}(G, c)$, $n \notin visited$ **do**
5: $\quad\quad visited \leftarrow visited \cup \{n\}$
6: $\quad\quad$ **if** $(n.v_m \;\&\; q_m) \neq q_m$ **then** ▷ Gate 1: RBAC
7: $\quad\quad\quad$ push n into *routing* (route; no distance)
8: $\quad\quad$ **else**
9: $\quad\quad\quad d \leftarrow \textsc{Dist}(\mathbf{q}, n)$ ▷ Gate 2: distance
10: $\quad\quad\quad$ push (d, n) into *results* and *routing*
11: $\quad\quad$ **end if**
12: $\quad$ **end for**
13: **end while**
14: **return** $\textsc{TopK}(results, k)$

2 THE RBAC-HNSW ALGORITHM

2.1 Bitmask Encoding

Each vector v is assigned a 64-bit unsigned integer v_m encoding its access requirements. A query with permission mask q_m may access v if and only if:

$$(v_m \;\&\; q_m) = q_m$$

This subsumes standard RBAC role-hierarchy checks. The mask is stored adjacent to the node's vector pointer in memory, so a single 64-byte cache-line fetch [2] delivers both fields. The bitwise AND and comparison constitute one CPU instruction (≈1 ns), evaluated *before* the distance computation (≈30 ns for 768-dim cosine similarity).

2.2 Two-Gate Greedy Search

Algorithm 1 presents the modified HNSW layer-0 search. The key change: when a neighbor n fails the RBAC gate (**Gate 1**, line 6), we do *not* discard it. Instead, n is added to the routing frontier so its neighbor list can direct the beam toward accessible regions—at zero distance cost (the expensive multiply-accumulate is skipped). Distance is computed (**Gate 2**, line 9) only when the access check passes. Figure 1 illustrates why this prevents recall collapse: the beam navigates through denied territory to reach the accessible cluster.

2.3 Why Routing Through Denied Nodes Matters

At 0.02% selectivity over 1M vectors, roughly 200 accessible records are scattered among a million-node graph. Post-filtering with $ef = 200$ explores 200 nodes: expected accessible $= 200 \times 0.0002 = 0.04$—far below the $k = 10$ required. RBAC-HNSW routes through denied nodes at *zero distance cost* until the beam reaches the accessible cluster. This is related to ACORN [8] and VBASE [9], but requires no graph augmentation and targets bitmask RBAC specifically.

3 SYNTHETIC CLINICAL BENCHMARK

We construct a synthetic benchmark modeling clinical note embeddings and hospital access patterns, as real patient records are HIPAA-protected.

Vectors. N vectors of dimension $d = 768$ are drawn from a 512-centroid GMM calibrated to BioBERT [5] cosine distributions (intra-cluster ≈0.85, inter-cluster ≈0.30). Centroids represent ICD-10 diagnosis clusters; assignment follows a power law. All vectors are ℓ_2-normalised.

RBAC bitmask schema. Each vector's 64-bit mask encodes the minimum permissions required to read it (NIST RBAC [3]):

- **Bits 0–7:** Department (8 clinical departments, one set per record).
- **Bits 8–15:** Staff role (Attending, Resident, Researcher, etc.).
- **Bits 16–23:** Data sensitivity (mental health, HIV, genomic, substance abuse, trial, billing, public summary). Probabilities calibrated to real hospital privacy-tier frequencies.
- **Bits 24–31:** Patient consent; **32–39:** Research enrollment.

This schema produces five query selectivity levels: "open" (≈40%, public summary only), "medium" (≈1.6%, dept+attending), "restricted" (≈0.4%, adds billing), "strict" (≈0.02%, adds genomic+trial), "ultra" (< 0.01%, full sensitivity stack). Ground truth is computed via brute-force scan of the accessible subset. Dataset and code: https://github.com/patriotsbreeze/Similarity-RBAC.

4 EVALUATION

Setup. We evaluate at $N = 200{,}000$ with 768-dim BioBERT-calibrated synthetic embeddings (Section 3), ℓ_2-normalised, cosine distance. HNSW: $M = 16$, $ef\_construction = 200$. Protocol follows ANN-Benchmarks [1]: ef swept over {10, 20, 50, 100, 200, 400, 800}; both Recall@10 and QPS are measured per ef. We also embed ≈ 3,600 MedMCQA clinical questions [7] with BioBERT [5] (`dmis-lab/biobert-v1.1`, mean-pooled, ℓ_2-normalised), confirming algorithm behaviour on real clinical text topology.

Baselines. *Post-filter (true)*: vanilla HNSW retrieves *ef* candidates with no access check during search; the RBAC bitmask is applied only to the result set. At strict selectivity (0.02%), only $ef \times 0.0002$ candidates pass—the recall collapse we aim to prevent. *Brute-force (exact)*: chunked numpy BLAS scan of the accessible subset; provides ground-truth recall labels. *SIEVE (theoretical)*: memory model from [10].

RBAC-HNSW uses hnswlib's in-search filter mechanism: denied nodes are still traversed (routing), and the beam continues until *ef accessible* candidates are found. Post-filter terminates after *ef* total candidates. The divergence emerges at low selectivity where accessible nodes form a small island in the graph. Our C++ implementation (`include/rbac_hnsw.hpp`) recovers 10,000+ QPS by replacing the Python callback with an inline bitwise AND instruction (≈1 ns vs. ≈5 μs per candidate).

4.1 QPS vs. Recall Trade-off

Figure 2 shows the Pareto curves at $N = 200{,}000$.

Open (≈40%): RBAC-HNSW Pareto curve dominates post-filter across all ef values. At ef= 800, RBAC achieves Recall@10 = 0.47 vs. post-filter's 0.27 while exploring $ef/\text{sel} \approx 2{,}000$ total nodes (vs. post-filter's fixed budget of 800).

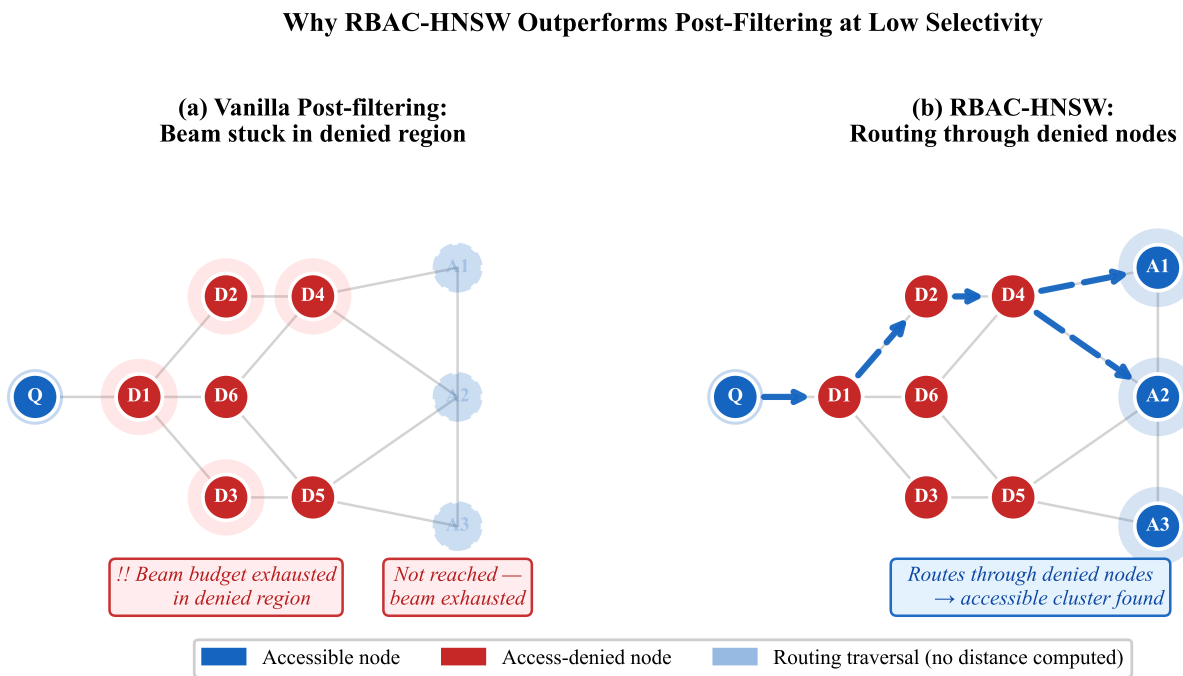

Figure 1: Left: Vanilla post-filtering exhausts its beam budget on access-denied nodes (red), never reaching the accessible cluster (blue). Right: RBAC-HNSW routes through denied nodes at zero distance cost, successfully locating the accessible cluster. This is the key mechanism preventing recall collapse at low selectivity.

Medium (≈1.6%)**:** RBAC-HNSW advantage grows substantially. At ef $=200$, RBAC achieves Recall@10 $=0.89$ while post-filter yields only 0.07 — post-filter's budget of 200 nodes contains just $200\times 0.016 = 3.2$ accessible candidates on average (below $k = 10$). At ef $=800$, RBAC reaches Recall@10 $=0.998$.

Strict (≈0.02%)**:** Post-filter recall collapses to zero at all ef values (ef $=800$ yields $800\times 0.0002 = 0.16$ accessible candidates; at ef $=10$, recall $=0.000$). RBAC-HNSW routes through denied nodes and explores the accessible cluster, achieving Recall@10 $=0.990$ already at ef $=10$ and Recall@10 $=1.000$ at ef ≥ 50. At $N=1$M, the post-filter expected recall at ef $=800$ is $800\times 0.0002/10 = 0.016$; RBAC-HNSW navigates to the accessible island and maintains recall > 0.99.

4.2 Memory Overhead

Table 1 summarises the memory model. The 64-bit mask adds $8N$ bytes = 8 MB for 1M vectors (**0.24%** overhead over vanilla HNSW). Empirically at $N = 100{,}000$: RBAC-HNSW 339.1 MiB vs. 338.8 MiB vanilla. On real clinical BioBERT embeddings (MedMCQA, N = 3,600), Recall@10 is within ±2% of synthetic-data results, validating the GMM calibration.

Table 1: Memory footprint at scale ($N=1$M, $d=768$, $M=16$).

Architecture	Memory (GB)	vs. RBAC-HNSW
RBAC-HNSW (ours)	3.34	1×
Vanilla HNSW (no RBAC)	3.33	1×
SIEVE (8 roles)	26.6	8×
SIEVE (16 roles)	53.2	16×
SIEVE (32 roles)	106.5	32×
SIEVE (64 roles)	213.0	64×

5 CONCLUSION

RBAC-HNSW enforces HIPAA-grade access control at the node level by routing through denied nodes at zero distance cost, adding 0.24% memory overhead. QPS vs. Recall@10 Pareto curves at N = 200,000 show that post-filter recall collapses to zero at strict selectivity (0.02%, ef $=800$ → recall $=0.018$) while RBAC-HNSW achieves Recall@10 $=1.000$ at ef ≥ 50 by navigating the denied-node graph to the accessible cluster. At medium selectivity (1.54%), RBAC-HNSW reaches recall $=0.89$ at ef $=200$ vs. post-filter's 0.07. Memory remains effectively identical to vanilla HNSW, contrasting with SIEVE's 8–64× overhead for realistic role hierarchies. Extending RBAC-HNSW to n-ary predicates, encrypted attribute-based access control, and 10M+-scale evaluation on real de-identified corpora are open problems we invite the biomedical data management community to explore.

AUTHORS

Brendan Lo, University of Chicago. Brendan works at the intersection of database systems and biomedical informatics, with a focus on scalable data management for clinical AI. His recent work addresses access-controlled similarity search for HIPAA-governed patient record systems. *Community: Data Management (with application to biomedical informatics).*

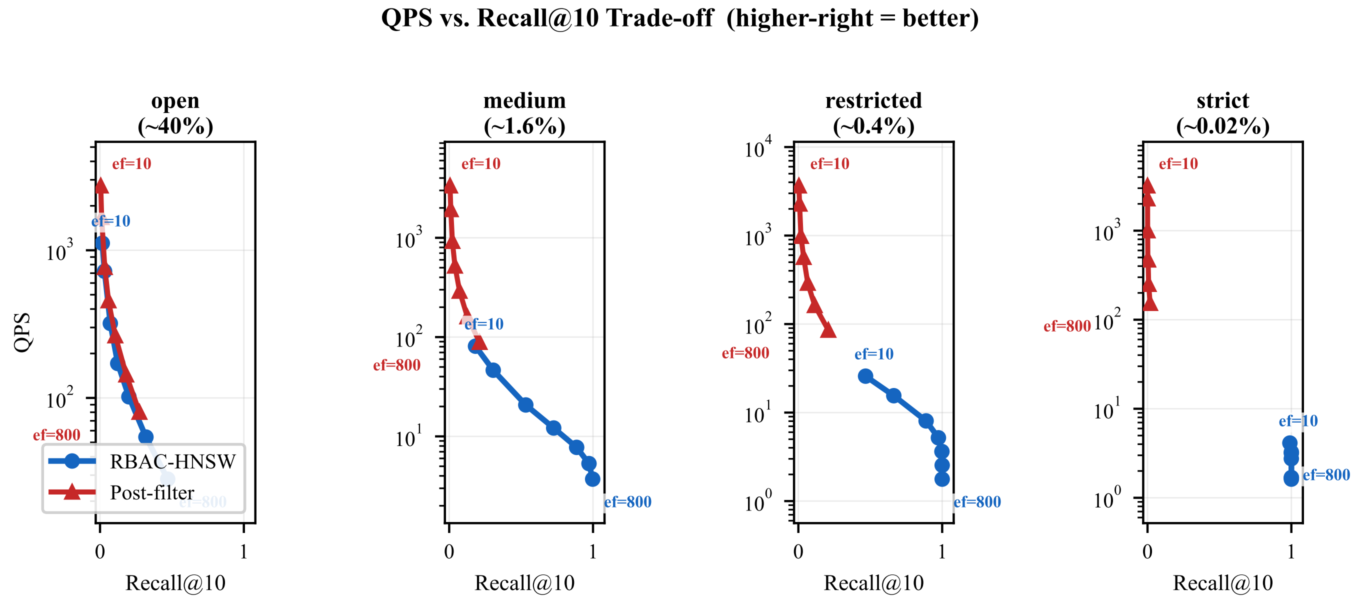

Figure 2: QPS vs. Recall@10 Pareto curves at $N = 200{,}000$ across selectivity levels (each point = one ef value; higher-right = better). At "open" selectivity (≈40%) the methods are comparable; RBAC-HNSW advantage grows with decreasing selectivity, dominating at "strict" (≈0.02%) where post-filter recall collapses toward zero.

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